

Model Practical

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SET-5

1. **Aim** :- To classify iris flower into their respective species using Sepal and petal measurements with machine learning.

Algorithm :-

1. Import required libraries.
2. Load the iris dataset.
3. visualize data using scatter plot.
4. split the dataset into training and testing.
5. Train a machine learning model.
6. predicts species for new test data.

Program :-

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import
train_test_split
data = pd.read_csv("IRIS.csv")
plt.xlabel("sepal width")
plt.ylabel("sepal length")
plt.legend()
plt.show()
print("predictes species:", prediction)
```

output :-

predicted Species: ['setosa']

Result :- The iris flower classification model was successfully implemented and correctly predicted as flower species based on input measurement.

2. Aim :- To find most specific hypothesis using the find s- algorithm from the given training dataset.

Algorithm :-

1. Initialize hypothesis with most specific values.
2. consider only positive examples.
3. Generalize hypothesis when attribute mismatch occur.
4. output final hypothesis.
5. Display the final hypothesis and accuracy values.

Program :-

```
import pandas as pd
data= [
    ['circular', 'large', 'Light', 'smooth', 'Thick', 'yes']
    ['circular', 'large', 'Light', 'Irregular', 'Thick', 'yes']
    ['oval', 'large', 'Light', 'Irregular', 'Thick', 'yes']
]
columns= ['shape', 'size', 'colour', 'surface']
if row ['target'] == 'yes':
    for i in range (len(hypothesis))
        elif hypothesis [i] != row[i]
```

Output :-

most specific hypothesis ['?', 'large', 'Light']

Result :-

The find s algorithm successfully generated the most specific hypothesis consistent with positive training.

Aim :- To implement Logistic Regression for classification of the iris dataset and evaluate its performance.

Algorithm:-

- Import libraries.
- Load dataset.
- Split data into training and testing sets
- Train Logistic Regression model
- predict output
- calculate accuracy.

Program :-

```
import pandas as pd
from sklearn.model_selection import
train-test-split
from sklearn.linear_model import
data = pd.read ("IRIS.csv")
accuracy = accuracy_score(y-test, y-pred)
y-pred = model.predict(x-test)
print ("Accuracy:", accuracy)
```

Output :-

Accuracy : 0.97.

Result :-

Logistic Regression classified the iris dataset with high accuracy, proving its effectiveness for linearly separable data.

1. **Aim** :- To implement the Naive Bayes algorithm for classification and evaluate its performance on the Iris dataset.

Algorithm :-

1. Import required Libraries.
2. Load dataset.
3. Split data into training and testing sets.
4. Train Naive Bayes classifier.
5. Predict test data.
6. Calculate accuracy.

Program :-

```
import pandas as pd
train, test = train_test_split(
    accuracy_score
    data = pd.read_csv("IRIS.csv")
    x = data.drop('species', axis=1)
    y = data['species']
    print("Accuracy", accuracy)
```

Output :-

Accuracy = 0.96

Result :-

The Naive Bayes algorithm was successfully implemented and achieved high classification accuracy and simplicity.